

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Period: \_\_\_\_\_

**Keep in Mind**

**This is a practice test.** It mirrors the real Unit 6 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

**Part A — Vocabulary (8 pts)**

Write the letter of the best description next to each term.

- |                              |                                                                                                                                         |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1. Index _____               | (A) Everything written underneath the radical symbol; when the index is even, it may not be negative.                                   |
| 2. Radicand _____            | (B) The operation that feeds one function's output into another function's input, written $(f \circ g)(x)$ .                            |
| 3. Principal root _____      | (C) The small number in the crook of the radical symbol that says which power is being undone.                                          |
| 4. Conjugate _____           | (D) The point $(h, k)$ that positions a transformed radical graph — an endpoint when the index is even, a turning point when it is odd. |
| 5. Extraneous solution _____ | (E) A candidate that satisfies the powered-up equation but fails when it is tested in the original.                                     |
| 6. Composition _____         | (F) The non-negative value the radical symbol by itself returns; a $\pm$ must be written in by hand.                                    |
| 7. Inverse function _____    | (G) The function that reverses another's inputs and outputs; its graph is the mirror image over the line $y = x$ .                      |
| 8. Anchor point _____        | (H) A binomial with the same two terms but the opposite middle sign; multiplying by it turns an irrational denominator rational.        |

**Part B — Multiple Choice (12 pts)**

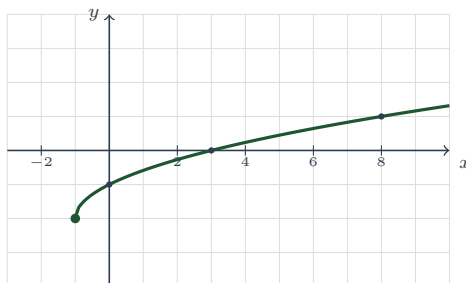
Circle the best answer.

- Written with a rational exponent,  $\sqrt[3]{x^5}$  is (A)  $x^{5/3}$  (B)  $x^{3/5}$  (C)  $x^{15}$  (D)  $x^{-3/5}$
- The domain of  $f(x) = \sqrt{7-2x}$  is (A)  $x \geq \frac{7}{2}$  (B)  $x \geq -\frac{7}{2}$  (C)  $x \leq \frac{7}{2}$  (D) all real numbers
- The graph of  $y = \sqrt{3x-12} + 5$  begins at the point (A) (12, 5) (B) (4, -5) (C) (-4, 5) (D) (4, 5)
- In simplest form,  $\sqrt{75} + \sqrt{12}$  is (A)  $\sqrt{87}$  (B)  $7\sqrt{3}$  (C)  $7\sqrt{6}$  (D)  $10\sqrt{3}$
- Which equation has *no real solution*?  
(A)  $\sqrt[3]{x+1} = -4$  (B)  $\sqrt{x-1} = 7$  (C)  $\sqrt{x+6} = -2$  (D)  $\sqrt{x} = 0$
- If  $f(x) = \sqrt{x}$  and  $g(x) = x + 9$ , then  $(f \circ g)(x) =$  (A)  $\sqrt{x} + 9$  (B)  $\sqrt{9x}$  (C)  $x\sqrt{x} + 9\sqrt{x}$  (D)  $\sqrt{x+9}$

**Part C — Short Answer & Computation (40 pts)**

Show your work. Assume every variable is non-negative unless told otherwise.

1. **Read the graph.** The radical function  $f$  is graphed below; each gridline is one unit and the solid dot is the endpoint.



- (a) Coordinates of the endpoint: \_\_\_\_\_
- (b) Domain (interval notation): \_\_\_\_\_; range: \_\_\_\_\_
- (c)  $x$ -intercept: \_\_\_\_\_;  $y$ -intercept: \_\_\_\_\_
- (d) Interval(s) where  $f$  is increasing: \_\_\_\_\_
- (e) Does  $f$  have an absolute maximum or an absolute minimum? Give it and say where it occurs.  
\_\_\_\_\_
- (f) End behavior: as  $x \rightarrow \infty$ ,  $f(x) \rightarrow$  \_\_\_\_\_; as  $x \rightarrow -\infty$ , \_\_\_\_\_
- (g) Interval(s) where  $f(x) < 0$ : \_\_\_\_\_

2. **Radicals and rational exponents.** Rewrite or evaluate. Leave no negative exponents.

- (a) Write  $\sqrt[4]{x^3}$  with a rational exponent. \_\_\_\_\_
- (b) Write  $y^{5/2}$  as a radical. \_\_\_\_\_
- (c)  $27^{2/3} =$  \_\_\_\_\_ (d)  $16^{-1/2} =$  \_\_\_\_\_
- (e) Write  $x^{1/2} \cdot x^{1/3}$  as a single radical. \_\_\_\_\_

3. **Simplify. Add or subtract where you can.**

(a)  $\sqrt{72}$  (b)  $\sqrt[3]{54}$

(c)  $\sqrt{20x^3}$  (d)  $5\sqrt{12} - \sqrt{27}$

4. **Multiply, then rationalize.** Every answer must be in simplest form.

(a)  $\sqrt{6} \cdot \sqrt{15}$  (b)  $(2 + \sqrt{5})(2 - \sqrt{5})$

(c)  $\frac{6}{\sqrt{3}}$  (d)  $\frac{4}{3 - \sqrt{5}}$

5. **Transformations.** Let  $g(x) = -2\sqrt{x+3} + 4$ .

- (a) List every transformation that takes  $y = \sqrt{x}$  to  $g$ . \_\_\_\_\_
- (b) Anchor point: \_\_\_\_\_
- (c) Domain: \_\_\_\_\_; range: \_\_\_\_\_
- (d) Is the anchor an absolute maximum or an absolute minimum? \_\_\_\_\_
- (e) A different graph of the form  $y = \sqrt{x-h} + k$  begins at  $(5, -2)$ . Write its equation. \_\_\_\_\_
- (f) For  $p(x) = \sqrt[3]{x-1} + 2$ , give the anchor point \_\_\_\_\_ and the domain \_\_\_\_\_. In one sentence, say why that domain is not like  $g$ 's.
- \_\_\_\_\_

6. **Solve. Test every candidate in the original equation.**

(a)  $\sqrt{2x+3} = x$

(b)  $\sqrt[3]{x-4} = -2$

(c)  $\sqrt{3x-2} + 4 = 1$

(d)  $x^{2/3} = 25$

7. **Composition.** Let  $f(x) = \sqrt{x}$  and  $g(x) = x - 6$ .

- (a)  $(f \circ g)(x) =$  \_\_\_\_\_, domain: \_\_\_\_\_
- (b)  $(g \circ f)(x) =$  \_\_\_\_\_, domain: \_\_\_\_\_
- (c) Are these the same function? Answer using *both* the rule and the domain.
- \_\_\_\_\_

(d)  $(f \circ g)(10) =$  \_\_\_\_\_

(e) For  $p(x) = 2x + 1$  and  $q(x) = x^2 - 3$ , find  $(p \circ q)(-2)$ . \_\_\_\_\_

8. **Inverses.**

- (a)  $f(x) = 4x - 7$ . Find  $f^{-1}(x)$ . \_\_\_\_\_
- (b) Verify by composition — *both* orders — that your answer really is the inverse.

(c)  $g(x) = \sqrt{x+2}$ . Find  $g^{-1}(x)$  and state its domain.

(d) The point  $(7, 3)$  lies on the graph of  $g$ . What point must lie on the graph of  $g^{-1}$ ? \_\_\_\_\_

### Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The full analysis.** Consider  $f(x) = \sqrt{x-1} + 2$ . **(a)** Give the anchor point, the domain, and the range, both in interval notation. **(b)** Find the  $x$ -intercept and the  $y$ -intercept, or explain why one does not exist. **(c)** List the transformations that take  $y = \sqrt{x}$  to  $f$ . **(d)** Find  $f^{-1}(x)$  and state its domain, saying exactly where that restriction comes from. **(e)** Justify with composition — both orders — that  $f$  and  $f^{-1}$  are inverses. **(f)** A classmate writes  $y = (x-2)^2 + 1$  with *no* restriction and calls it the inverse. Explain why that is not the inverse of  $f$ .

2. **Modeling.** Ignoring friction, a roller-coaster car that starts from rest and drops  $h$  feet reaches the bottom at a speed of

$$v(h) = 8\sqrt{h} \text{ feet per second.}$$

- (a)** State the domain that makes sense here, and connect it to the rule about what an even index will accept. **(b)** Find  $v(9)$  and  $v(36)$ , with units. What does quadrupling the drop do to the speed, and which feature of  $\sqrt{\cdot}$  is responsible? **(c)** A car reaches 56 feet per second at the bottom. Find the height of the drop algebraically, and check your answer. **(d)** Write the inverse,  $h$  as a function of  $v$ , and say what it tells an engineer that  $v(h)$  does not. **(e)** Part (c) required squaring both sides. Explain why there is no danger of an extraneous solution in this problem.